# IGCSE 0607 Practice Questions — E3.4 — Combined Questions & Markscheme

## Finding a quadratic function using given information

**Total Questions:** 30 — **Total Marks:** 156 Marks

---

## Table of Contents & Quick Navigation


- **E3.4** Finding a quadratic function using given information
  → **Paper 2 (Non-Calculator)** *(16 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*


---

**1** A quadratic function has $x$-intercepts at $x = -3$ and $x = 4$.
The coefficient of $x^2$ is 1.

Write down the equation of this function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[2]**

**2** A quadratic function has $x$-intercepts at $x = 4$ and $x = -2$.
The coefficient of $x^2$ is 1.

Find the equation of the quadratic function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

**3** A quadratic function $f(x)$ has $x$-intercepts at $x = -4$ and $x = 3$.
The coefficient of $x^2$ is 1.

Find the equation of the function in the form $f(x) = x^2 + bx + c$.

$f(x) =$ ..... **[2]**

**4** A quadratic function has the equation $y = x^2 + bx + c$.
The vertex of the graph of this function is $(4, -3)$.

Find the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[3]**

**5** A quadratic function is given by

$$y = x^2 + bx + c$$

The vertex of the graph of this function is $(3, -4)$.

Find the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[3]**

**6** A quadratic function has a minimum point at $(3, -4)$ and passes through the point $(5, 0)$.
The coefficient of $x^2$ is 1.

Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[3]**

**7** The graph of a quadratic function, $y = x^2 + bx + c$, crosses the $x$-axis at $(-4, 0)$ and $(2, 0)$.

**(a)** Work out the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[2]**

**(b)** Find the coordinates of the vertex of the graph.

( ..... , ..... ) **[2]**

**8** A quadratic function $y = f(x)$ has its vertex at $(3, -4)$.
The coefficient of $x^2$ is 1.

**(a)** Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

**(b)** Find the coordinates of the points where the graph of $y = f(x)$ crosses the $x$-axis.

..... **[2]**

**9** A quadratic function is given by the equation $y = (x - h)^2 + k$.
The vertex of the graph of this function is $(3, -2)$.

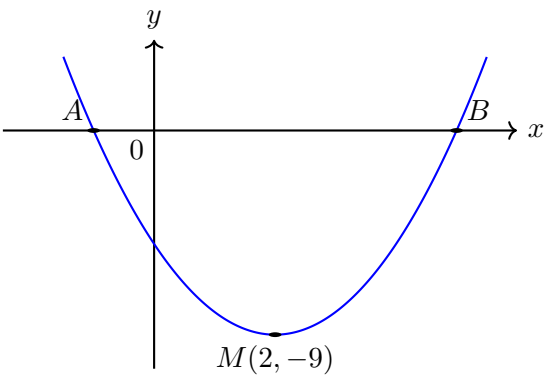
**(a)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

**(b)** The graph passes through the point $(5, p)$.
Work out the value of $p$.

$p =$ ..... **[2]**

**10** The diagram shows the graph of a quadratic function $y = x^2 + px + q$.
The graph has a minimum point (vertex) at $M(2, -9)$.




**(a)** Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[3]**

**(b)** The graph crosses the $x$-axis at the points $A$ and $B$.
Find the coordinates of $A$ and the coordinates of $B$.

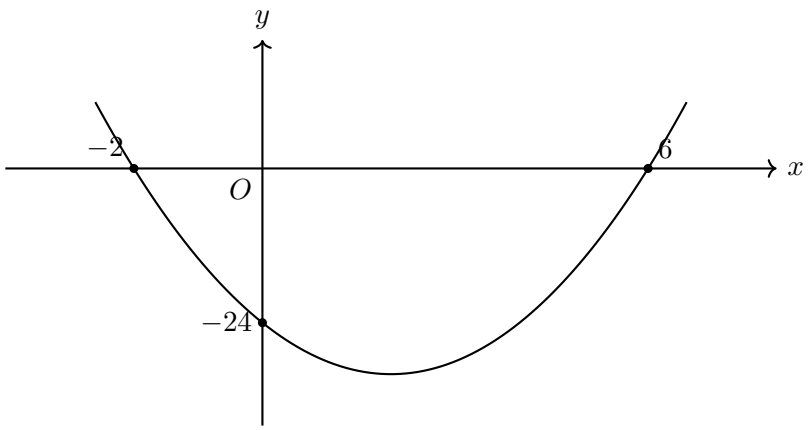
$A =$ ..... , $B =$ ..... **[3]**

**11** The graph of a quadratic function $y = f(x)$ intersects the $x$-axis at $(-2, 0)$ and $(5, 0)$.
The graph intersects the $y$-axis at $(0, -20)$.

Work out the function $f(x)$, giving your answer in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

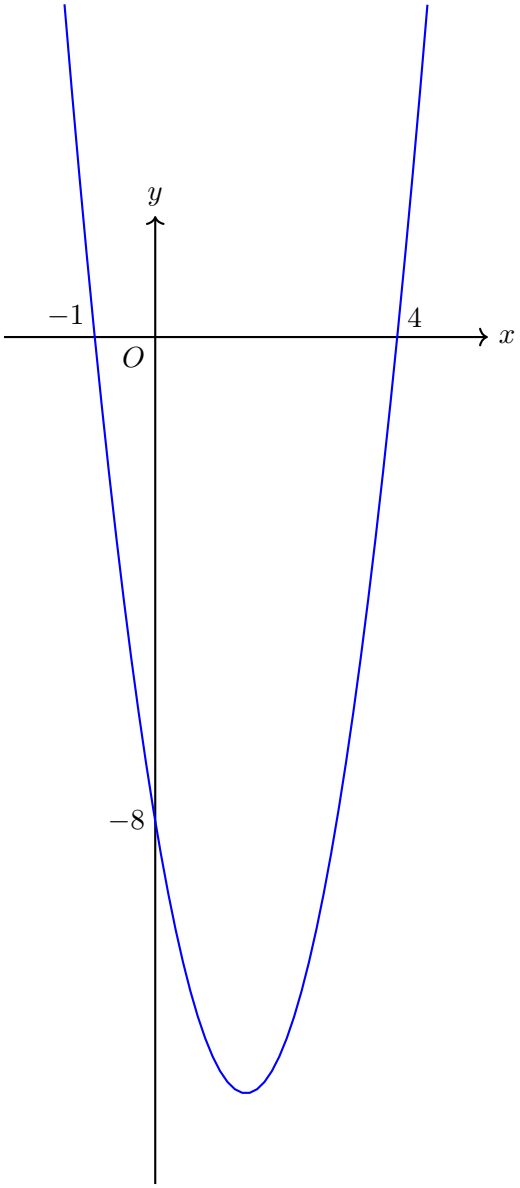
**12** The diagram shows a sketch of the quadratic graph $y = f(x)$.
The graph crosses the $x$-axis at $(-2, 0)$ and $(6, 0)$.
It crosses the $y$-axis at $(0, -24)$.




Find the expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**13** The diagram shows a sketch of the quadratic function $y = f(x)$.
The graph crosses the $x$-axis at $(-1, 0)$ and $(4, 0)$.
The graph crosses the $y$-axis at $(0, -8)$.




**(a)** Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

**(b)** Find the coordinates of the minimum point of the graph.

( ..... , ..... ) **[3]**

**14** A quadratic function is given by $f(x) = a(x-h)^2 + k$.
The graph of $y = f(x)$ has a maximum point at $(-1, 8)$ and passes through the point $(2, -10)$.

**(a)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

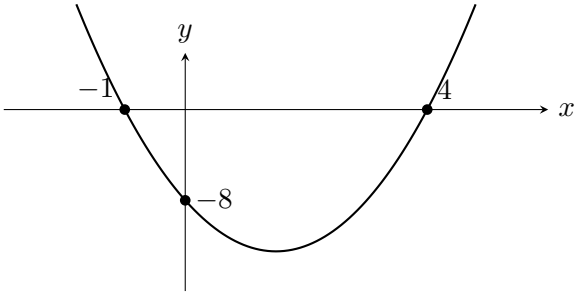
**(b)** Find the value of $a$.

$a =$ ..... **[3]**

**(c)** Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[3]**

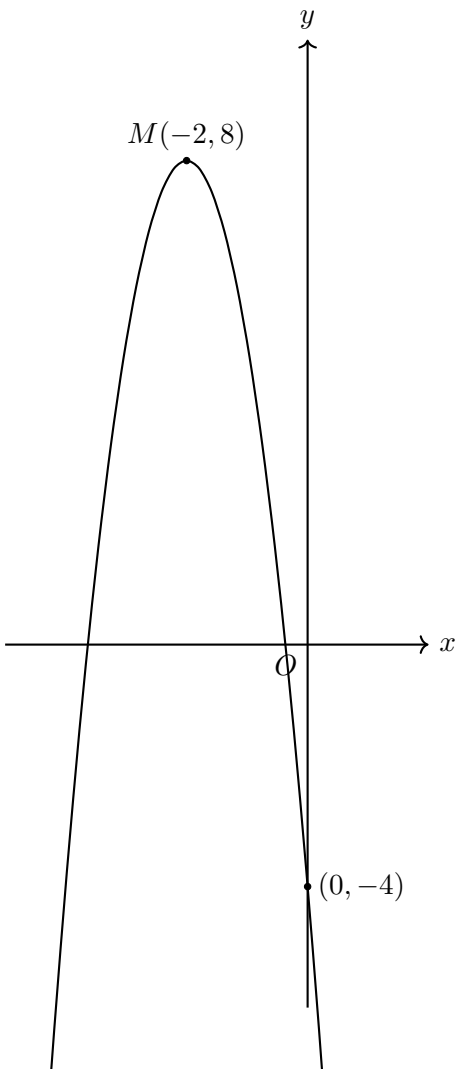
**15** The diagram shows the graph of a quadratic function $y = f(x)$.
The graph crosses the $x$-axis at $x = -1$ and $x = 4$.
The graph crosses the $y$-axis at $(0, -8)$.




Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**16** The diagram shows a sketch of the curve $y = f(x)$ where $f(x)$ is a quadratic function.
The maximum point on the curve is $M(-2, 8)$ and the curve crosses the $y$-axis at $(0, -4)$.




**(a)** Find the equation of the curve.
Give your answer in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**(b)** Solve $f(x) = 0$ to find the exact $x$-coordinates of the points where the curve crosses the $x$-axis.

Give your answers in the form $\frac{p \pm q\sqrt{6}}{r}$, where $p$, $q$ and $r$ are integers.

$x =$ ..... **[3]**

**17** A quadratic curve has its vertex at $(3, -5)$ and passes through the point $(1, 7)$.
Find the equation of the curve in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

**18** The graph of a quadratic function has its vertex at $(-2, 5)$ and passes through the point $(1, -13)$. Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

**19** The graph of a quadratic function has its vertex at $(2, 7)$ and passes through the point $(4, -5)$.

**(a)** Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

**(b)** Find the $y$-intercept of the graph.

..... **[1]**

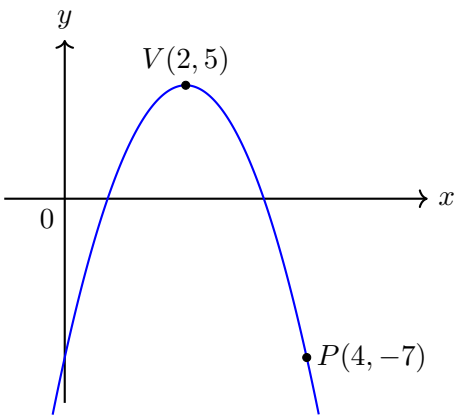
**20** A quadratic curve crosses the $x$-axis at $(-1, 0)$ and $(5, 0)$.

The curve also passes through the point $(2, -18)$.

Find the equation of the curve in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

**21** The diagram shows a sketch of the graph of a quadratic function $y = a(x - h)^2 + k$.
The vertex of the graph is $V(2, 5)$.
The graph passes through the point $P(4, -7)$.




**(a)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

**(b)** Work out the value of $a$.

$a =$ ..... **[2]**

**(c)** Find the coordinates of the $y$-intercept of the graph.

( ..... , ..... ) **[2]**

**22** The graph of a quadratic function $f(x)$ intersects the $x$-axis at $x = -2$ and $x = 5$.
The graph passes through the point $(3, 10)$.

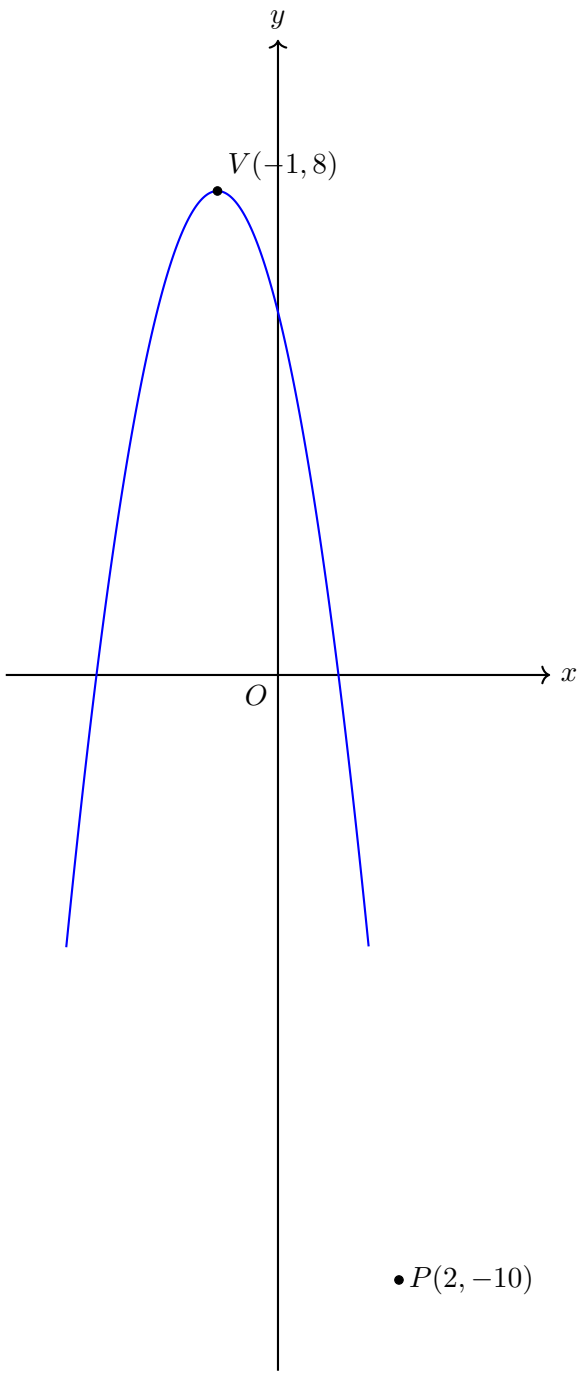
**(a)** Find an expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**(b)** Find the coordinates of the vertex of the graph of $y = f(x)$.

( ..... , ..... ) **[2]**

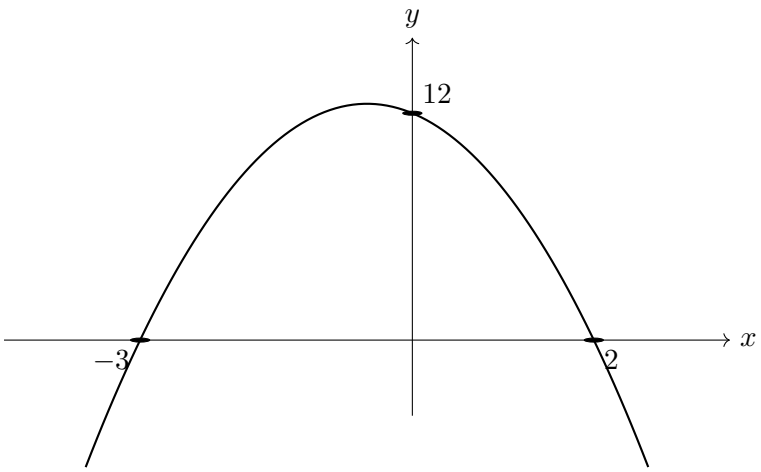
**23** The diagram shows a sketch of the graph of $y = ax^2 + bx + c$.
The vertex of the graph is $V(-1, 8)$ and it passes through the point $P(2, -10)$.




Work out the values of $a$, $b$ and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

**24** The diagram shows the sketch of a quadratic function.
The curve crosses the $x$-axis at $-3$ and 2, and intersects the $y$-axis at 12.



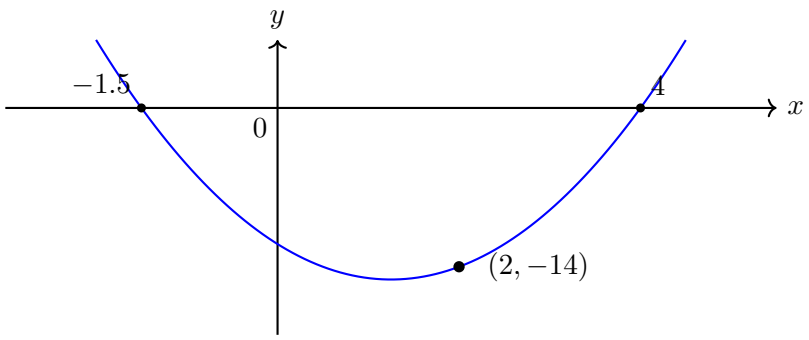

**(a)** Find the equation of the quadratic function.
Give your answer in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

**(b)** Calculate the coordinates of the vertex of this curve.

( ( ..... , ..... ) ) **[2]**

**25** The diagram shows a sketch of the quadratic function $y = f(x)$.
The curve crosses the $x$-axis at $x = -1.5$ and $x = 4$.
It passes through the point $(2, -14)$.




**(a)** Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**(b)** Find the exact coordinates of the turning point of the graph.

( ..... , ..... ) **[3]**

**26** A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The maximum point of the graph of $y = f(x)$ is $(-4, 20)$.
The graph intersects the $x$-axis at $x = -2$ and $x = p$.

**(a)** **(i)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

**(ii)** Find the value of $a$.

$a =$ ..... **[2]**

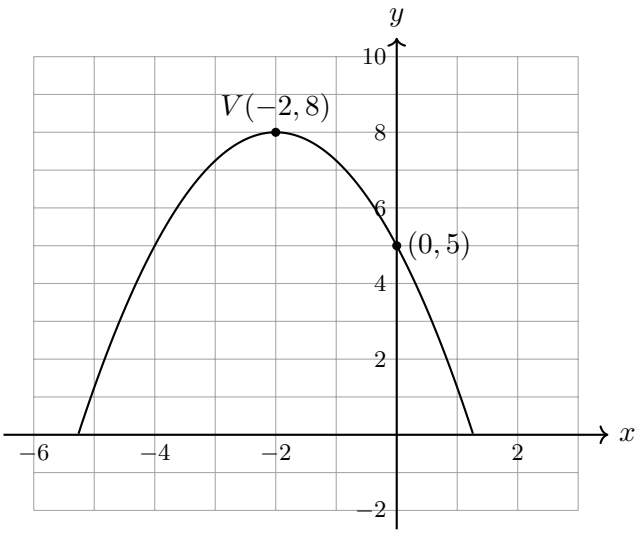
**(iii)** Find the value of $p$.

$p =$ ..... **[1]**

**(iv)** Express $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

**27** The diagram shows the graph of a quadratic function $y = f(x)$.
The vertex of the graph is at $V(-2, 8)$ and it intersects the $y$-axis at $(0, 5)$.




**(a)** Find the equation of the function $f(x)$ in the form $f(x) = a(x - h)^2 + k$.

$f(x) =$ ..... **[3]**

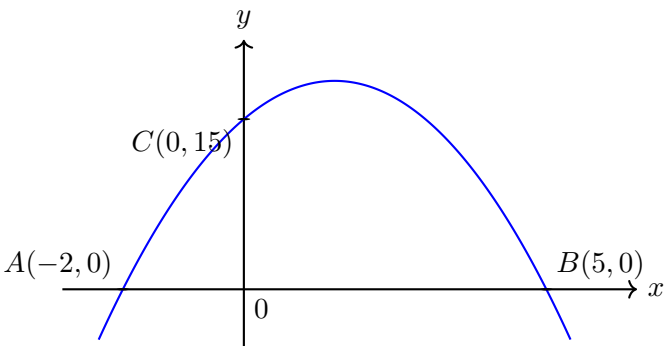
**(b)** Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

**(c)** Calculate the exact $x$-coordinates of the points where the graph intersects the $x$-axis.
Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[3]**

**28** The diagram shows a sketch of the quadratic curve $y = ax^2 + bx + c$.
The curve crosses the $x$-axis at $A(-2, 0)$ and $B(5, 0)$, and crosses the $y$-axis at $C(0, 15)$.




**(a)** Find the values of $a$, $b$, and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

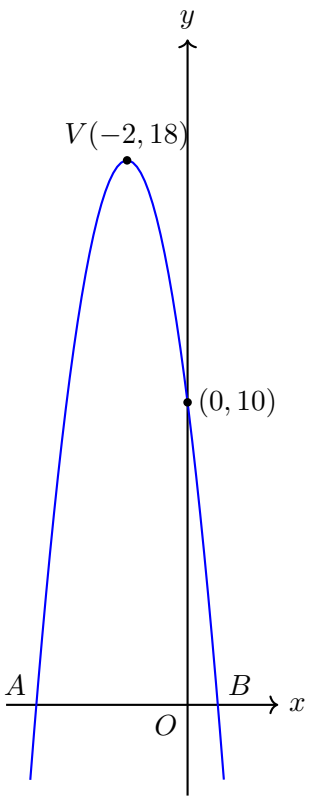
**(b)** Find the coordinates of the vertex of the curve.

( ..... , ..... ) **[3]**

**(c)** The equation $ax^2 + bx + c = k$ has no real roots.
Write down the inequality for $k$.

..... **[1]**

**29** The diagram shows a sketch of the graph of a quadratic function $y = f(x)$.
The maximum point of the graph is $V(-2, 18)$.
The graph crosses the $y$-axis at $(0, 10)$.




**(a)** Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

**(b)** The graph crosses the $x$-axis at $A$ and $B$.
Find the length of the line segment $AB$.

$AB =$ ..... **[3]**

**(c)** The line $y = -2x + k$ is a tangent to the curve $y = f(x)$.

Find the value of $k$.

$k =$ ..... **[4]**

**30** The quadratic function $f(x) = ax^2 + bx + c$ has a minimum turning point at $(1, -8)$.
The graph of $y = f(x)$ crosses the $x$-axis at $x = -1$ and $x = 3$.

**(a)** Show that $f(x) = 2x^2 - 4x - 6$.

**[3]**

**(b)** Solve the equation.

$f(x) = 10$

$x =$ ..... , $x =$ ..... **[3]**

**(c)** A straight line has equation $y = mx - 8$.
The line is a tangent to the graph of $y = f(x)$.

Find the possible values of $m$.

$m =$ ..... , $m =$ ..... **[4]**

**(d)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ k \end{pmatrix}$.

The new graph has no $x$-intercepts.

Write down the inequality for $k$.

..... **[2]**

**1** $y = (x+3)(x-4)$ **[M1]**

$x^2 - x - 12$ **[A1]**

[Back to Question] *[Total Marks: 2]*

**2** $(x-4)(x+2)$ **[M1]**

$x^2 - 2x - 8$ **[A1]**

[Back to Question] *[Total Marks: 2]*

**3** Uses $(x-p)(x-q)$ with $p = -4$ and $q = 3$: $(x+4)(x-3)$ **[M1]**

$x^2 + x - 12$ **[A1]**

[Back to Question] *[Total Marks: 2]*

**4** Uses the vertex form $y = (x-4)^2 - 3$ **[M1]**

Expands to $x^2 - 8x + 16 - 3$ **[M1]**

$b = -8$ and $c = 13$ **[A1]**

[Back to Question] *[Total Marks: 3]*

**5** Uses the vertex form $y = (x-3)^2 - 4$ OR uses $-\dfrac{b}{2(1)} = 3$ **[M1]**

$b = -6$ **[A1]**

$c = 5$ **[A1]**

[Back to Question] *[Total Marks: 3]*

**6** $y = (x-3)^2 - 4$ **[M1]**

$y = x^2 - 6x + 9 - 4$ **[M1]**

$x^2 - 6x + 5$ **[A1]**

[Back to Question] *[Total Marks: 3]*

**7 (a)** $x^2 + bx + c = (x+4)(x-2)$ **[M1]**

$b = 2$, $c = -8$ **[A1]**

**(b)** Finds $x$-coordinate of vertex: $x = \dfrac{-4+2}{2} = -1$ **[M1]**

$(-1, -9)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**8** **(a)** Uses vertex form: $y = (x-3)^2 - 4$ **[M1]**

$x^2 - 6x + 5$ **[A1]**

**(b)** Sets $x^2 - 6x + 5 = 0 \implies (x-1)(x-5) = 0$ **[M1]**

$(1, 0)$ and $(5, 0)$ **[A1]**

**[Back to Question]** *[Total Marks: 4]*

**9** **(a)** $h = 3$ **[B1]**

$k = -2$ **[B1]**

**(b)** $p = (5-3)^2 - 2$ **[M1]**

$p = 2$ **[A1]**

**[Back to Question]** *[Total Marks: 4]*

**10** **(a)** Uses vertex form: $y = (x-2)^2 - 9$ or $x$-coordinate of vertex $\dfrac{-p}{2} = 2$ **[M1]**

Expands to find $p$ and $q$, or solves $2^2 + 2p + q = -9$ **[M1]**

$p = -4$ and $q = -5$ **[A1]**

**(b)** $x^2 - 4x - 5 = 0$ **[M1]**

$(x-5)(x+1) = 0$ **[M1]**

$A = (-1, 0)$ and $B = (5, 0)$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**11** $f(x) = a(x+2)(x-5)$ **[M1]**

$a(0+2)(0-5) = -20$ or $-10a = -20$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 20$ **[A1]**

**[Back to Question]** *[Total Marks: 4]*

**12** $f(x) = a(x+2)(x-6)$ **[M1]**

$-24 = a(0+2)(0-6)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 24$ **[A1]**

**[Back to Question]** *[Total Marks: 4]*

**13** **(a)** Uses intercept form: $y = a(x+1)(x-4)$ **[M1]**

Substitutes $(0, -8)$ to find $a$: $-8 = a(0+1)(0-4) \implies -4a = -8 \implies a = 2$ **[M1]**

$2x^2 - 6x - 8$ **[A1]**

**(b)** Finds the $x$-coordinate of the vertex by symmetry: $x = \frac{-1+4}{2} = 1.5$ (or uses $x = -\frac{b}{2a}$) **[M1]**

Substitutes $x = 1.5$ into the equation: $y = 2(1.5)^2 - 6(1.5) - 8$ **[M1]**

$(1.5, -12.5)$ (or $(\frac{3}{2}, -\frac{25}{2})$) **[A1]**

[Back to Question] *[Total Marks: 6]*

**14 (a)** $h = -1$ **[B1]**

$k = 8$ **[B1]**

**(b)** $-10 = a(2 - (-1))^2 + 8$ **[M1]**

$9a = -18$ **[M1]**

$-2$ **[A1]**

**(c)** $-2(x + 1)^2 + 8$ **[M1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$-2x^2 - 4x + 6$ **[A1]**

[Back to Question] *[Total Marks: 8]*

**15** $f(x) = a(x + 1)(x - 4)$ **[M1]**

$-8 = a(0 + 1)(0 - 4)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 8$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**16 (a)** Forms the equation $f(x) = a(x + 2)^2 + 8$ **[M1]**

Substitutes $(0, -4)$ into their equation:

$-4 = a(0 + 2)^2 + 8$ **[M1]**

$4a = -12 \implies a = -3$ **[A1]**

$-3x^2 - 12x - 4$ **[A1]**

**(b)** Sets $-3(x + 2)^2 + 8 = 0$ or $-3x^2 - 12x - 4 = 0$ **[M1]**

Rearranges to $(x + 2)^2 = \frac{8}{3}$ OR correctly substitutes into the quadratic formula:

$x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(-3)(-4)}}{2(-3)}$ **[M1]**

$\frac{-6 \pm 2\sqrt{6}}{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**17** $y = a(x-3)^2 - 5$ **[M1]**

$7 = a(1-3)^2 - 5 \implies 12 = 4a \implies a = 3$ **[M1]**

$3(x-3)^2 - 5$ **[A1]**

**[Back to Question]** *[Total Marks: 3]*

**18** $y = a(x+2)^2 + 5$ **[M1]**

$-13 = a(1+2)^2 + 5$ **[M1]**

$-18 = 9a$

a = -2$[M1]$

$-2(x+2)^2 + 5$ **[A1]**

**[Back to Question]** *[Total Marks: 3]*

**19** **(a)** Uses vertex form: $y = a(x-2)^2 + 7$ **[M1]**

Substitutes $(4, -5)$ into their equation: $-5 = a(4-2)^2 + 7$ **[M1]**

$-3(x-2)^2 + 7$ **[A1]**

**(b)** $-5$ **[B1]**

**[Back to Question]** *[Total Marks: 4]*

**20** Writes the function in the factored form $y = a(x+1)(x-5)$ **[M1]**

Substitutes $(2, -18)$ into their equation:

$-18 = a(2+1)(2-5)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 10$ **[A1]**

**[Back to Question]** *[Total Marks: 4]*

**21** **(a)** $h = 2$ **[B1]**

$k = 5$ **[B1]**

**(b)** Substitutes $(4, -7)$ into equation: $-7 = a(4-2)^2 + 5$ **[M1]**

$a = -3$ **[A1]**

**(c)** Substitutes $x = 0$ into $y = -3(x-2)^2 + 5$ **[M1]**

$(0, -7)$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**22** **(a)** Sets up the factored form $f(x) = a(x+2)(x-5)$ **[M1]**

Substitutes the point $(3, 10)$: $10 = a(3+2)(3-5)$ **[M1]**

Solves for $a$: $10 = a(5)(-2) \implies a = -1$ **[A1]**

$f(x) = -x^2 + 3x + 10$ **[A1]**

**(b)** Finds the $x$-coordinate of the vertex (e.g., midpoint of intercepts $\frac{-2+5}{2} = 1.5$ or $x = \frac{-b}{2a}$) **[M1]**

$(1.5, 12.25)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**23** $y = a(x+1)^2 + 8$ **[M1]**

$-10 = a(2+1)^2 + 8$ **[M1]**

$9a = -18 \implies a = -2$ **[A1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$a = -2$, $b = -4$, $c = 6$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**24 (a)** $y = a(x+3)(x-2)$ **[M1]**

$12 = a(0+3)(0-2)$ **[M1]**

$12 = -6a$

a = -2[$M1$]

$-2x^2 - 2x + 12$ **[A1]**

**(b)** $x = \frac{-3+2}{2}$ or $x = \frac{-(-2)}{2(-2)} = -0.5$ **[M1]**

$y = -2(-0.5)^2 - 2(-0.5) + 12 = 12.5$ **[M1]**

$(-0.5, 12.5)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**25 (a)** Sets up equation using $x$-intercepts: $f(x) = a(x+1.5)(x-4)$ OR $f(x) = a(2x+3)(x-4)$ **[M1]**

Substitutes $(2, -14)$ to find $a$: $-14 = a(2+1.5)(2-4)$ **[M1]**

Solves for $a$: $a = 2$ **[A1]**

$2x^2 - 5x - 12$ **[A1]**

**(b)** Identifies $x$-coordinate of vertex as midpoint of roots: $x = \frac{-1.5+4}{2} = 1.25$ OR uses $x = \frac{-b}{2a}$ **[M1]**

Substitutes $x = 1.25$ into their function: $y = 2(1.25)^2 - 5(1.25) - 12$ **[M1]**

$(1.25, -15.125)$ OR $\left(\frac{5}{4}, -\frac{121}{8}\right)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**26** **(a)** **(i)** $h = -4$ **[B1]**

$k = 20$ **[B1]**

**(ii)** $0 = a(-2+4)^2 + 20$ **[M1]**

$a = -5$ **[A1]**

**(iii)** $-6$ **[B1]**

**(iv)** $f(x) = -5(x+4)^2 + 20 = -5(x^2 + 8x + 16) + 20$ **[M1]**

$-5x^2 - 40x - 60$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**27** **(a)** Uses the vertex coordinates to write $f(x) = a(x+2)^2 + 8$ **[M1]**

Substitutes the $y$-intercept $(0, 5)$: $5 = a(0+2)^2 + 8$ **[M1]**

$f(x) = -0.75(x+2)^2 + 8$ or equivalent using fractions like $-\frac{3}{4}$ **[A1]**

**(b)** Expands $(x+2)^2$ as $x^2 + 4x + 4$ **[M1]**

$f(x) = -0.75x^2 - 3x + 5$ **[A1]**

**(c)** Equates function to zero: $-0.75(x+2)^2 + 8 = 0$ or $-0.75x^2 - 3x + 5 = 0$ **[M1]**

Rearranges to $(x+2)^2 = \frac{32}{3}$ or correctly applies the quadratic formula **[M1]**

$x = 1.27$ and $x = -5.27$ **[A1]**

[Back to Question] *[Total Marks: 8]*

**28** **(a)** $c = 15$ **[B1]**

$y = a(x+2)(x-5)$ **[M1]**

$15 = a(0+2)(0-5)$ **[M1]**

$a = -1.5$ **[A1]**

$b = 4.5$ **[A1]**

**(b)** $x = \frac{-2+5}{2} = 1.5$ **[M1]**

$y = -1.5(1.5)^2 + 4.5(1.5) + 15$ **[M1]**

$(1.5, 18.375)$ or $\left(\frac{3}{2}, \frac{147}{8}\right)$ **[A1]**

**(c)** $k > 18.375$ **[B1]**

[Back to Question] *[Total Marks: 9]*

**29** **(a)** Uses vertex form: $y = a(x+2)^2 + 18$ **[M1]**

Substitutes $(0, 10)$ to find $a$: $10 = a(0+2)^2 + 18 \implies 4a = -8 \implies a = -2$ **[M1]**

$-2x^2 - 8x + 10$ **[A1]**

**(b)** Sets $f(x) = 0$: $-2x^2 - 8x + 10 = 0$ **[M1]**

$x^2 + 4x - 5 = 0 \implies (x+5)(x-1) = 0 \implies x = -5$ or $x = 1$ **[M1]**

6 **[A1]**

**(c)** Equates line and curve: $-2x^2 - 8x + 10 = -2x + k$ **[M1]**

Rearranges to form a quadratic equation: $2x^2 + 6x + (k-10) = 0$ (or equivalent) **[M1]**

Sets discriminant $\Delta = 0$ for a tangent: $6^2 - 4(2)(k-10) = 0$ **[M1]**

14.5 (or $\frac{29}{2}$) **[A1]**

**[Back to Question]** *[Total Marks: 10]*

**30** **(a)** Forms equation using $x$-intercepts: $f(x) = a(x+1)(x-3)$ OR using vertex $f(x) = a(x-1)^2 - 8$ **[M1]**

Substitutes appropriate point to find $a$: $a(1+1)(1-3) = -8 \implies -4a = -8 \implies a = 2$ **[M1]**

Expands to show final form: $2(x^2 - 2x - 3) = 2x^2 - 4x - 6$ **[A1]**

**(b)** $2x^2 - 4x - 6 = 10 \implies 2x^2 - 4x - 16 = 0$ **[M1]**

Factors correctly to $(x-4)(x+2) = 0$ OR uses quadratic formula correctly **[M1]**

$x = 4$ and $x = -2$ **[A1]**

**(c)** Sets equations equal: $2x^2 - 4x - 6 = mx - 8$ leading to $2x^2 - (4+m)x + 2 = 0$ **[M1]**

States condition for tangent (discriminant $= 0$): $(-(4+m))^2 - 4(2)(2) = 0$ **[M1]**

Simplifies to $(4+m)^2 = 16$ or $m^2 + 8m = 0$ **[M1]**

$m = 0$, $m = -8$ **[A1]**

**(d)** Recognises minimum $y$-value must be strictly greater than 0: $-8 + k > 0$ **[M1]**

$k > 8$ **[A1]**

**[Back to Question]** *[Total Marks: 12]*